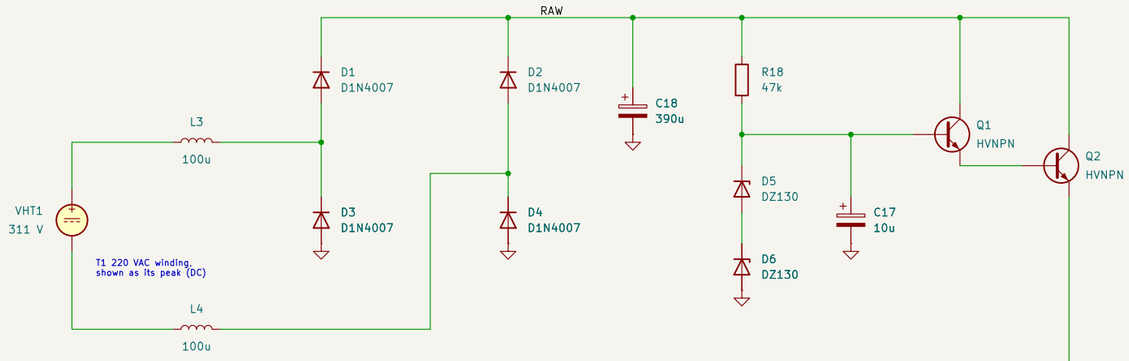
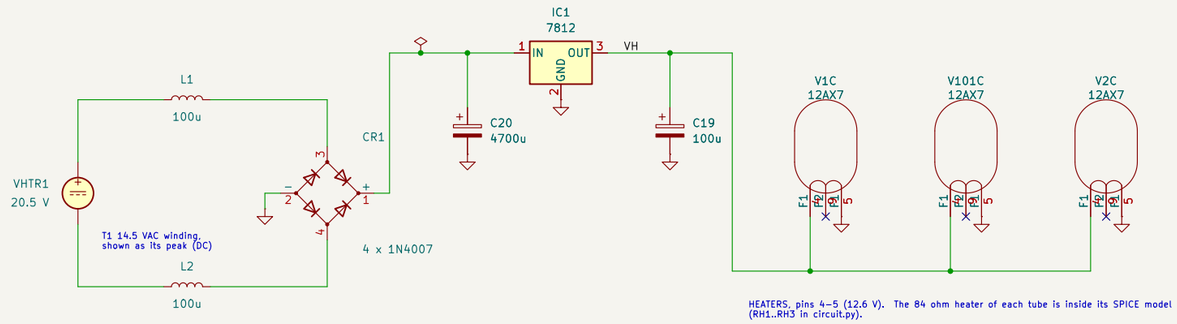
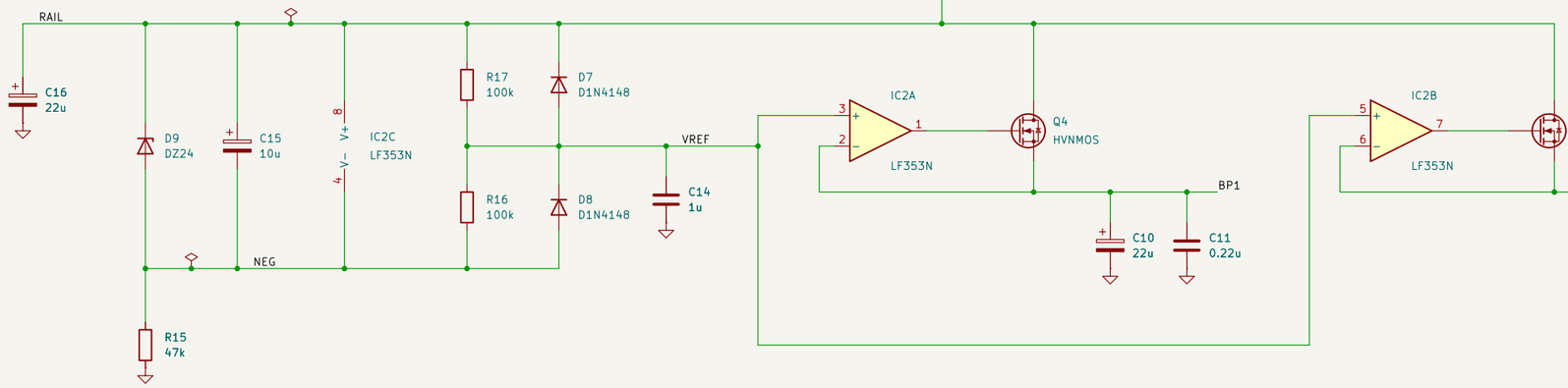
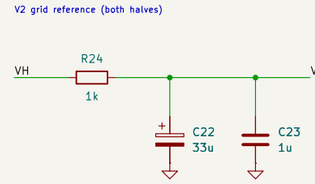
MOSCODE "SUPER IT" — NEW YORK AUDIO LABS Rev D — POWER SUPPLY (COMMON TO BOTH CHANNELS)

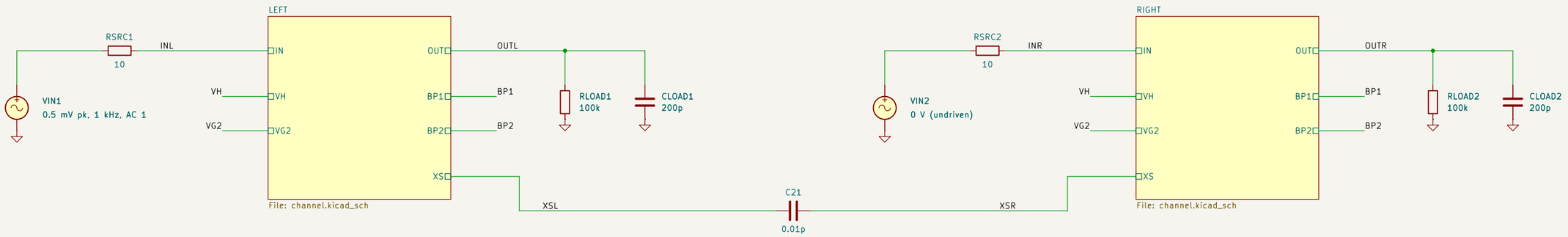
Where the values come from (hidden fields Status / Note on every part say more):
  K known (drawing / owners' notes): VHT VHTR RH1 RH2 RH3
  P read from board photographs: C18 R15 C14 IC2 Q3 C13 Q4 C11 R24 C22 C21
  E estimate: L3 L4 D1 D2 D3 D4 R18 D5 D6 C17 Q1 Q2 C16 D9 C15 R17 R16 D7 D8 C12 C10 L1 L2 CR1 C20 IC1 C19 C23
T1 (transformer box), fuses F1 / F2 and the mains side are not drawn; VHT1 / VHTR1 stand for the two secondaries.
IC2 is an LF353N (both sections follow the same filtered reference VREF; Q4 feeds stage 1, Q3 stage 2 + followers).












```
.ac dec 50 10 100k
.options reltol=1e-5 abstol=1e-12 vntol=1e-6
```

Simulation: the two lines above go into the netlist; the first is the analysis KiCad runs
(Inspect > Simulator, Run). Edit it, or set another analysis in the simulator, e.g.
    .tran 5u 20m 10m   1 kHz, 0.5 mV sine from VIN1 (left channel); plot V(/OUTL)
    .op   DC operating point
Useful signals: /OUTL /OUTR /INL /LEFT/p1 /LEFT/g6 /LEFT/p2 /LEFT/s5 /BP1 /BP2 /VREF

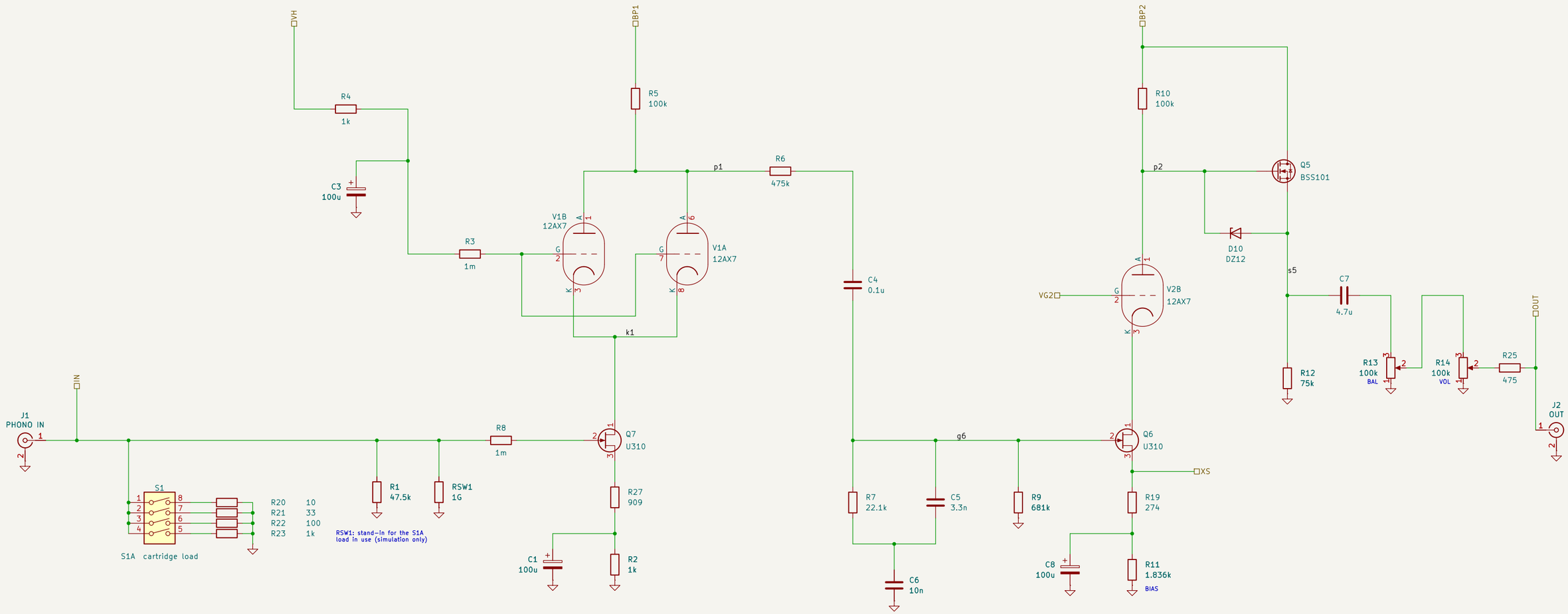
Simulate: Inspect > Simulator, then Run
Reconstructed from the Rev D scan; values from ../sim/circuit.py

Sheet: /
File: superit.kicad_sch

Title: Moscode "SuperIt" Rev D — power supply and test fixture

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.8		Id: 1/3

MOSCODE "SUPER IT" — NEW YORK AUDIO LABS Rev D — AUDIO CIRCUIT, ONE CHANNEL



Where the values come from (hidden fields Status / Note on every part say more):
  K known (drawing / owners' notes): V1 V2
  P read from board photographs: R1 R8 Q7 R27 R2 C1 R5 R3 R4 R6 C4 R7 C5 C6 R9 Q6 R19 R11 C8 R10 Q5 R12 R13 R14 R25
  E estimate: R20 R21 R22 R23 RSW C3 D10 C7

Q7 / Q6 are U310 metal-can JFETs (model: low-pinch-off corner of the data sheet, see superit.lib).
R3 and R8 are wire links on the photographed board (1 mOhm here so the netlist keeps them).
R13 / R14: pot wiper position is Sim.Params pos (1 = full level, 0 = off).
S1 and J1 / J2 are excluded from simulation; RSW1 stands in for the selected S1A load.
V1: both halves in parallel (pins 1,2,3 and 6,7,8). V2: this channel uses one half,
the other channel the other half (the stock ECC83 symbol letters the 1-2-3 half 'B').

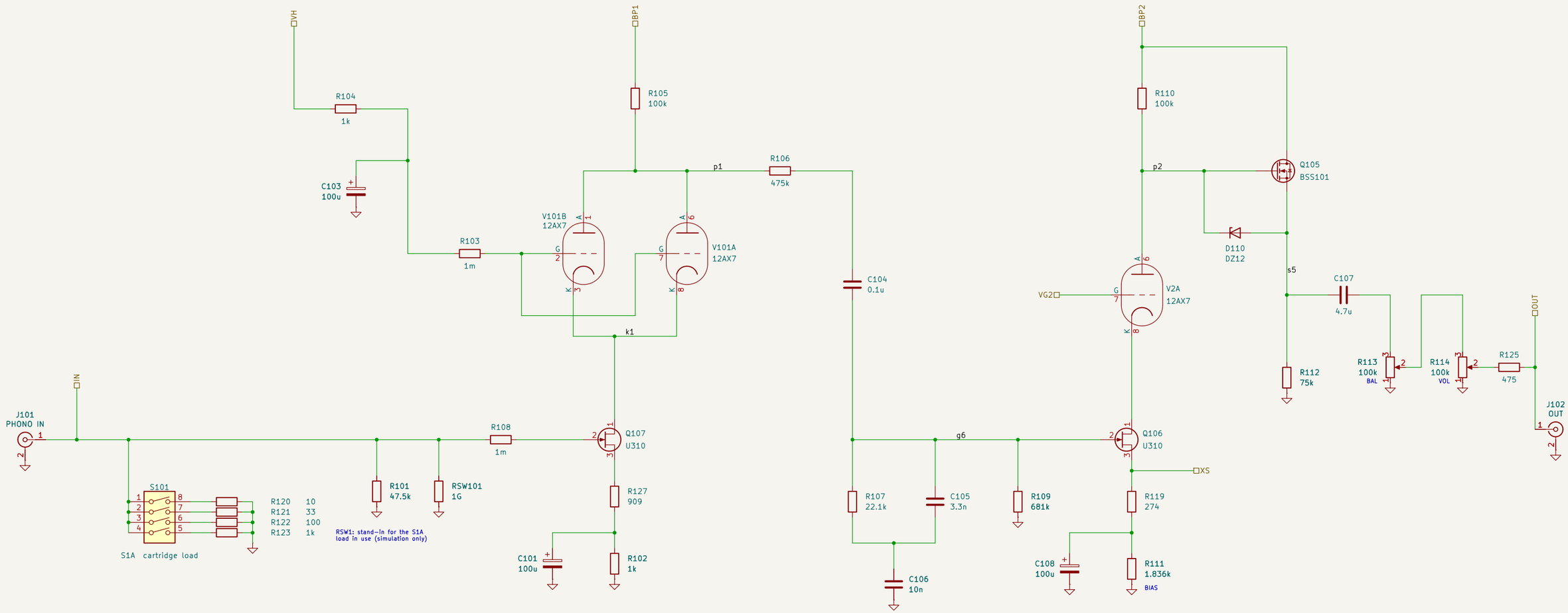
Values from ../sim/circuit.py — regenerate with gen_kicad.py
Sheet used twice (LEFT, RIGHT); right-channel refs are +100

Sheet: /LEFT/
File: channel.kicad_sch

Title: Moscode "SuperIt" Rev D — audio channel

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.8		Id: 2/3

MOSCODE "SUPER IT" — NEW YORK AUDIO LABS Rev D — AUDIO CIRCUIT, ONE CHANNEL



STAGE 1: JFET + 12AX7 cascode

PASSIVE RIAA

STAGE 2: JFET + 12AX7 cascode,
MOSFET follower

Where the values come from (hidden fields Status / Note on every part say more):
  K known (drawing / owners' notes): V1 V2
  P read from board photographs: R1 R8 Q7 R27 R2 C1 R5 R3 R4 R6 C4 R7 C5 C6 R9 Q6 R19 R11 C8 R10 Q5 R12 R13 R14 R25
  E estimate: R20 R21 R22 R23 RSW C3 D10 C7

Q7 / Q6 are U310 metal-can JFETs (model: low-pinch-off corner of the data sheet, see superit.lib).
R3 and R8 are wire links on the photographed board (1 mOhm here so the netlist keeps them).
R13 / R14: pot wiper position is Sim.Params pos (1 = full level, 0 = off).
S1 and J1 / J2 are excluded from simulation; RSW1 stands in for the selected S1A load.
V1: both halves in parallel (pins 1,2,3 and 6,7,8). V2: this channel uses one half,
the other channel the other half (the stock ECC83 symbol letters the 1-2-3 half 'B').

Values from ../sim/circuit.py — regenerate with gen_kicad.py
Sheet used twice (LEFT, RIGHT); right-channel refs are +100

Sheet: /RIGHT/
File: channel.kicad_sch

Title: Moscode "SuperIt" Rev D — audio channel

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.8		Id: 3/3